

Network Topology

What is Network Topology?

Network topology is the **physical or logical layout** of how computers, cables, and devices are connected.

It refers to how devices in a computer network are connected. Network Topology is the way that defines the structure, and how these components are connected.

It is divided into two types:

Physical Topology = Real connections (wires, hubs)

It indicates the arrangement of different elements of a network. It reflects the physical layout of devices and cables to form a connected network.

Logical Topology = How data flows between devices

It defines the path that data packets take as they travel through the network independent of its physical layout.

Types of Network Topology

1. Bus Topology

Definition: All devices are connected to a **single cable** (the bus).

📌 Also called **Linear Bus** if it has two endpoints.

Bus topology carries transmitted data through the cable because data reaches each node, the node checks the destination address (MAC/IP address) to determine if it matches their address. If the address does not match with the node, the node does nothing more. But if the addresses of nodes match to addresses contained within the data then they process knowledge. In the bus, communication between nodes is done through a network cable.

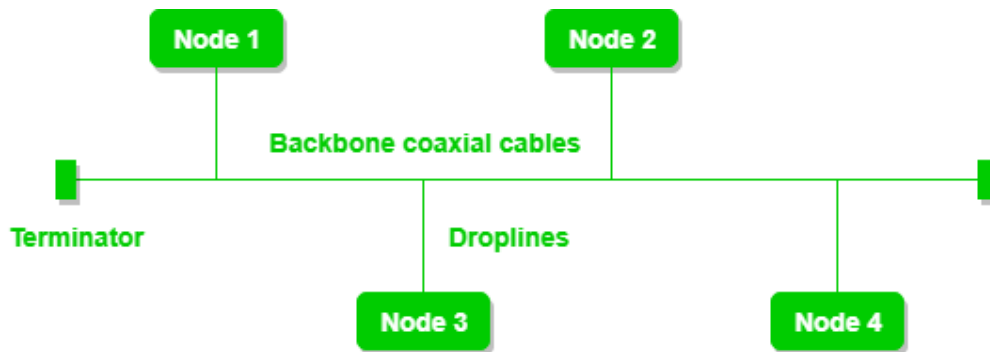
Advantages of Bus Topology

1. It works very efficiently well when there is a small network.
2. The length of cable required is less than a star topology.
3. It is easy to connect or remove devices in this network without affecting any other device.
4. Very cost-effective as compared to other network topology i.e. mesh and star
5. Easy to expand by joining the two cables together.

Disadvantages of Bus Topology

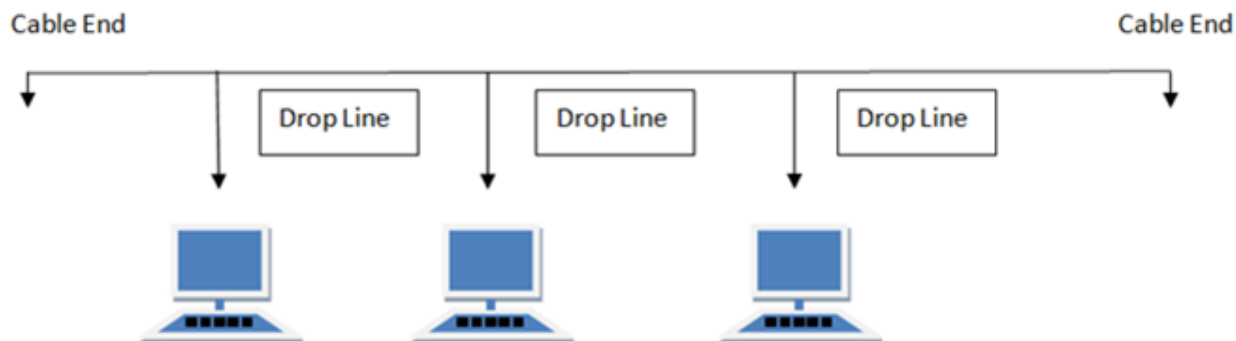
1. Bus topology is not good for large networks.
2. Identification of problems becomes difficult if the whole network goes down.
3. Troubleshooting individual device issues is very hard.

4. Additional devices slow the network down.
5. If the main cable is damaged, the whole network fails or splits into two.
One cable failure = Entire network fails
6. Poor performance with heavy traffic



Applications of Bus Topology:

1. Old office or school labs where all computers were connected along one main cable.
2. Building Automation Systems: Bus topology is employed in building automation and HVAC (heating, ventilation, and air conditioning) structures to attach sensors.
3. Broadcast Industry: Used in small TV or radio station setups for internal communication and broadcasting signal distribution.
4. Small local offices and event management companies use temporary bus topologies for quick network setups during exhibitions or product demos.
5. Public Announcement (PA) System in Schools
6. One track (like a cable) connects many stations (devices). A train (data) goes along the track and stops at the right station.



A **dropline** is the cable that connects a device (computer, printer, etc.) to the main backbone cable (also called the bus).

Imagine the main bus cable as a highway, and each dropline as a short exit road that connects a building (device) to the highway.

Data Transmission in Bus Topology

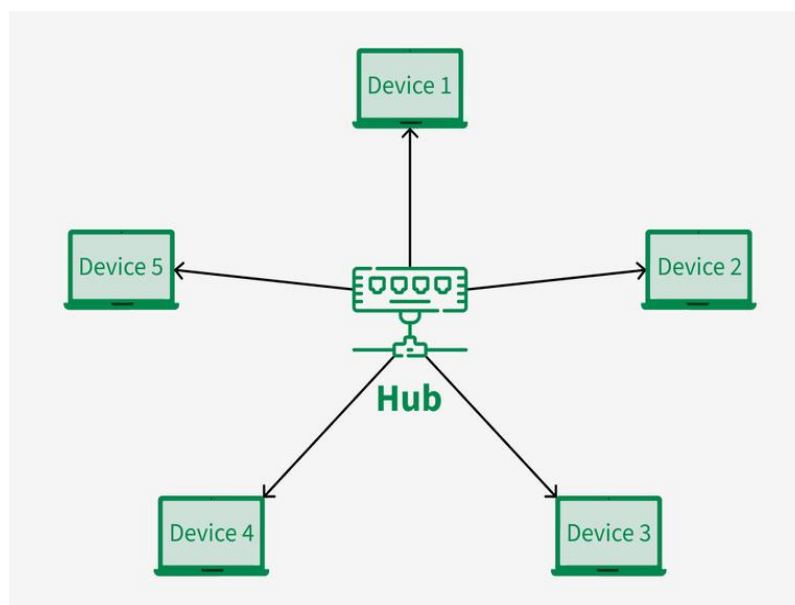
In a bus topology, data transmission takes place along a single backbone cable, known as the bus. The process of sending data in this particular network setup can be dissected in the following manner:

1. **Structure:** A single backbone cable links all devices (nodes) and serves as a shared communication medium.
2. **Data Transmission:** When a device needs to send data, it sends data bidirectionally along the bus. Data sent by any node is transmit to all other nodes in the network.
3. **Data Reception:** While transmission in bus topology, each node analyzes the data destination address to determine if it is the designated receiver. If the node's address matches the destination address, the data is processed by the node, Otherwise, the data is ignored.
4. **Collision Detection:** The bus being shared by all nodes increases the probability of data collisions when two devices transmit at the same time. In the majority of bus topologies, a technique such as Carrier Sense Multiple Access with Collision Detection ([CSMA/CD](#)) is implemented to handle collisions. [CSMA/CD](#) requires that devices supervise the bus to verify its availability before transmitting data and to recognize and handle collisions if they happen.

2. Star Topology

Star Topology is a network setup in which each device is connected to a central node called a hub. The hub manages the data flow between the devices. If one device wants to send data to another device, it has to first send the information to the hub, and then the hub transmits that data to the required device.

After receiving the data, the hub checks for the destination address, and if the address matches the data is accepted otherwise data is rejected.



A star may be a Local Area Network (LAN) topology where all nodes are individually connected to a central connection point (hub). The number of links required to connect nodes in the star topology is N where N is the number of nodes. The **hub or switch** manages and controls all functions of the network.

Types of Star Topology

- Active Star Topology
- Passive Star topology

Active Star Topology: In active star topology, the central hub regenerates the signal when it passes through it. This simply means that the hub not only works as a connector but also boosts the signal and makes sure the signals reach their destination properly.

Passive Star Topology: In passive star topology, the central hub simply connects the devices, but does not regenerate the signals. Here, the devices connected are responsible for the transmission of signals. Passive star topology is recommended for smaller setups.

Applications of Star Topology

- **Home Networks:** Star Topology is used in a home network where all devices connects to a connected to a central router that manages the Internet.
- **ATM Networks:** In banks, ATMs are also connected in as a star network, as it helps in managing data efficiently.
- **Hospital Networks:** Hospital devices, medicinal equipment are connected in the star topology. It helps in easily accessing data records.
- **CCTV Networks:** Star Topolgy are also used in CCTV Networks where multiple cameras are connected to central video recorder.

What is the Importance of Star Topology in LAN?

Star Topology is important in Local Area Network because it helps in easy management and scalability of the network. As central hub is present in the star topology, it becomes simple to monitor and control network traffic among devices present in the network. Because of star topology, adding or removing devices becomes easy and it also helps in expanding networks.

Advantages of Star Topology

- It is very reliable as if one cable or device fails then all the others will still work.
- It is high performing as no data collisions can occur.
- It is less expensive because each device only needs one I/O port and wishes to be connected to the hub with one link.
- Easy fault detection because the links are often easily identified.
- No disruptions to the network when connecting or removing devices.

- Each device requires just one port i.e. to attach to the hub.
- If N devices are connected to each other in star, then the amount of cables required to attach them is N. So, it's easy to line up.

Disadvantages of Star Topology

- Requires more cable than [bus topology](#).
- If the connecting network device (network switch) fails, the nodes attached are disabled and can't participate in network communication.
- More expensive than linear bus topology due to the value of the connecting devices (network switches).
- If the hub goes down everything goes down, none of the devices can work without the hub.
- Hub requires more resources and regular maintenance because it's the central system of Star.
- Extra hardware is required (hubs or switches) which adds to the cost.
- Performance is predicated on the one concentrator i.e. hub.

Difference Between Star and Bus Topology

Star Topology	Bus Topology
All devices are connected to a central hub	All devices are connected to a single central cable (called the bus)
If a device fails, only that device is affected, the rest of the network keeps working	If a device fails, the rest of the network keeps working
Requires more cable as compared to Bus Topology	Requires less cables as compared to Star Topology
Motly used for small to medium size networks	Mostly used for simple networks having fewer devices

3. Ring Topology

In this topology device connections create a circular data path. In this each device is



connected to with its exactly two neighboring devices, like points on a circle which forms like a ring structure.

Advantages of Ring topology :

- In this data flows in one direction which reduces the chance of packet collisions.
- In this topology additional workstations can be added after without impacting performance of the network.
- Equal access to the resources.
- There is no need of server to control the connectivity among the nodes in the topology.
- It is cheap to install and expand.
- Minimum collision.
- Speed to transfer the data is very high in this type of topology.
- Due to the presence of token passing the performance of ring topology becomes better than bus topology under heavy traffic.
- Easy to manage.
- Ring network is extremely orderly organized where every device has access to the token and therefore the opportunity to transmit.

Disadvantages of Ring topology :

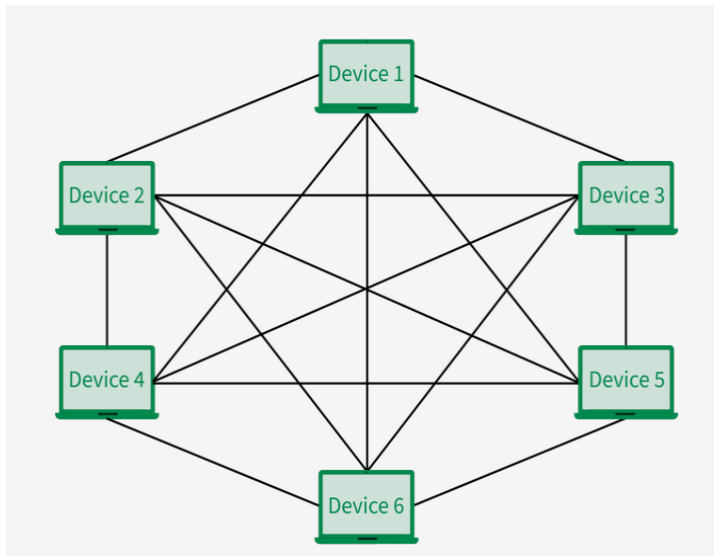
- Due to the Uni-directional Ring, a data packet (token) must have to pass through all the nodes.
- If one workstation shuts down, it affects whole network or if a node goes down entire network goes down.
- It is slower in performance as compared to the bus topology
- It is Expensive.
- Addition and removal of any node during a network is difficult and may cause issue in network activity.
- Difficult to troubleshoot the ring.
- In order for all the computer to communicate with each other, all computer must be turned on.
- Total dependence in on one cable.
- Not Scalable.

Application:

Manufacturing Industry: Used on factory floors where machines are interconnected in a ring for real-time data.

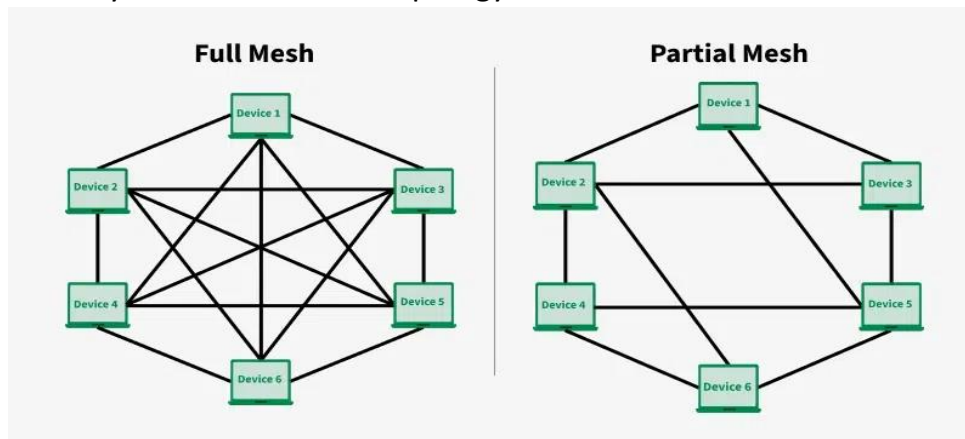
4. Mesh Topology

It is a network configuration where every device is interconnected with every other device, providing multiple route for data to travel. The nodes are connected to each other completely via a dedicated link during which information travels from nodes to nodes. If a mesh network has N nodes, then there are $N(N-1)/2$ links. Each computer not only sends its signals but also transfer data from other computers. The connections within the mesh are often wired or wireless. In this article, we will discuss the Mesh Topology in detail.



Types of Mesh Topology

- Fully-connected Mesh Topology
- Partially-connected Mesh Topology



Full Mesh Topology

All the nodes within the network are connected with every other. If there are n number of

nodes during a network, each node will have an $n-1$ number of connections. A full mesh provides an excellent deal of redundancy, but because it is expensive to implement, it's usually reserved for network backbones. Total number of links required for the mesh topology is $[n(n-1)]/2$.

Partial Mesh Topology

The partial mesh is more practical as compared to the full mesh. In a partially connected mesh, all the nodes aren't necessary to be connected with one another during a network. Peripheral networks are connected using partial mesh topology.

Applications of Mesh Topology

- Wireless Networks: Many homes and small office spaces use Mesh Wifi System for better internet coverage and connectivity.
- Industrial and Manufacturing Networks: Many industries use mesh topology to control machinery, ensuring better productivity.
- Smart Homes: Some advanced homes also use mesh topology for connecting home appliances like smart lights, security systems, etc.
- Military Communication: Mesh Topology are used in Military Purposes as if one connection fails, then there is always a chance of another connection.

Advantages of Mesh Topology

- In case of failure of a single device, the entire network didn't break.
- There is no traffic problem as there is a dedicated point to point links for every device.
- Mesh Topology provides high privacy and security.
- Data transmission is more consistent because failure doesn't disrupt its processes.
- Adding new devices won't disrupt transmission of data.

Disadvantages of Mesh Topology

- Mesh Topology is costly as compared to the other network topologies i.e. star, bus, point to point topology.
- Installation of nodes are difficult in mesh topology.
- Power requirement is higher as all the nodes will need to remain active all the time and share the load.
- Each node requires an extra utility cost.
- More maintenance is required in mesh topology.

Conclusion

In conclusion, mesh topology is a way to connect devices in a network where each device is linked directly to every other device. This setup is suitable for keeping the network reliable

because if one connection breaks, the data can still find another way to reach its destination. As Mesh Topology is expensive and complicated to set up as compared to other network types like [star topology](#), [bus topology](#), etc. But its advantages make it worth considering, especially in situations where constant communication is essential, like in military systems, industrial settings, or smart homes.

5. Hybrid topology

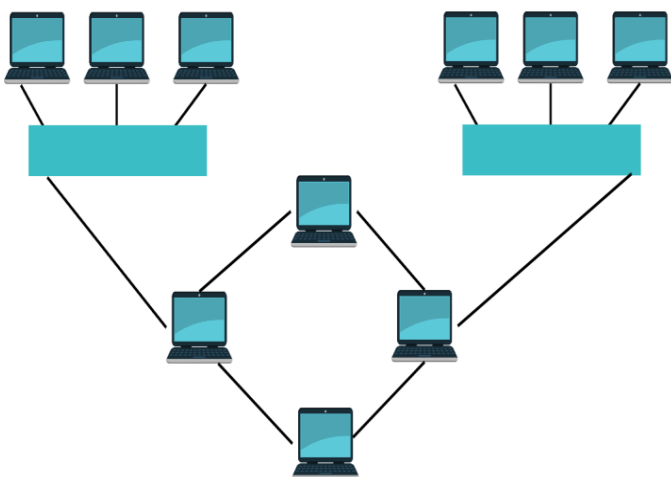
It is defined as a network topology that combines two or more different network topologies. A hybrid topology can be a combination of bus topology, ring topology and mesh topology. The selection of different types of network topologies combined together depends upon the number of computers, their location, and the required performance. In the hybrid topology network sections consist of a configuration of different types of network topologies. The structure of hybrid topology is more complex but offers various advantages such as flexibility and fault tolerance.

Working of Hybrid Topology

- Hybrid topology in computer networks is a combination of more than one topology.
- This hybrid topology makes use of standards such as Wi-Fi and Ethernet for performing its different operations.
- The functioning of hybrid topology depends upon various types of hybrid routers used such as switches and hubs as they can easily connect the devices that are connected over wired or wireless networks.
- The hybrid topology has different network branches and each branch as its own unique design.
- The functioning takes place by gaining various benefits of the both of the different topologies used.

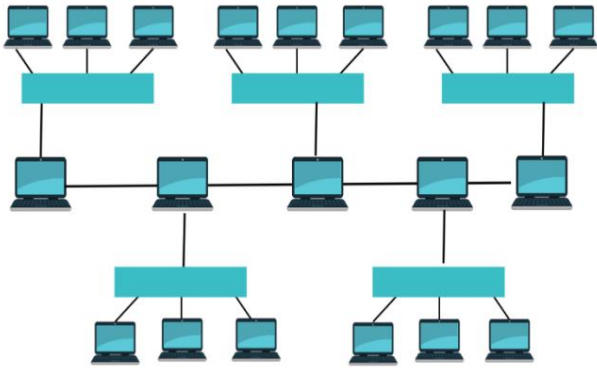
Types of Hybrid Topologies

1. Star-Ring Hybrid Topology



The combination of star and ring topology forms a star-ring topology. Two or more than two star topologies are connected together through a ring topology using a wired connection. The flow of data in star-ring topology is bidirectional or unidirectional. If any node of the original ring topology gets fail the bidirectional data flow provides with the surety that there will be no effect on the rest of the data in network flow.

2. Star-Bus Hybrid Topology



The combination of star and bus topology is known as star-bus hybrid topology. Two or more star topologies are connected together with the help of bus topology through a wired connection. The bus topology can interrelate different star topologies and offers with a backbone structure. The entire network is not affected in case of any node failure. The failed node can be then easily replaced and offers

with a easy way for adding or deleting the nodes. The overall network can be easily modified according to the need.

3. Hierarchical Network Topology

Hierarchical network topology is designed like a hierarchical tree and also known as network tree topology. Hierarchical network topology is a combination of star-ring hybrid topology and star-bus hybrid topology. The maximum level of hierarchical network topology is known as root node or parent node. The sub-nodes of parent node are known as child nodes. The last nodes that does does not have any child are known as leaf nodes.

Advantages of Hybrid Topology

- Adding a new node or deleting the existing node is easy in hybrid topologies.
- Hybrid topology is more secure, reliable, and scalable as compared to individual star, ring and mesh topology.
- Error detection and troubleshooting is easier in hybrid topology.
- When an organization has a large geographical area utilizing hybrid topology is considered as better option.
- Traffic with large volume is handled easily by the hybrid topology.
- The overall performance and speed is greater in hybrid topology.

Disadvantages of Hybrid Topology

- The design and implementation of hybrid network topology is difficult.
- More number of cables and other physical devices are required for hybrid topology.
- The process of installation of hybrid topology is difficult.
- The overall implementation, setup and process of hybrid topology is much more costlier.

Applications:

Large Enterprises: Combine topologies to fit different departmental needs.

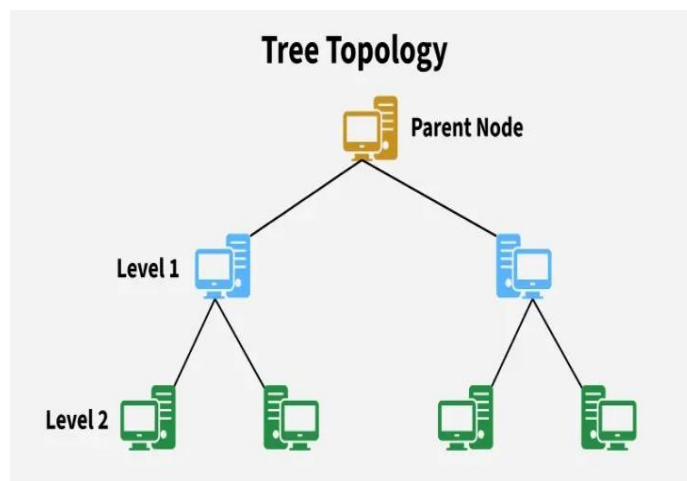
Universities: Different campuses or departments use different topologies that are integrated.

Tata Consultancy Services (TCS) and Infosys: Use hybrid topologies across large campus networks.

NASA: Integrates mesh, star, and ring topologies depending on the operation (satellite control vs. administrative networks).

6.Tree Topology

It has a root node and all other nodes are linked to it creating a hierarchy. It is also called “hierarchical topology”. It must have a minimum of three levels to the hierarchy.



- It merges the features of both [star](#) and [bus topologies](#). It includes a node known as the root that connects to one or more star networks called branches. Each branch can further extend into sub-branches, forming a structure resembling that of a tree.
- Tree topology is hierarchical, allowing for easy expansion of the network and better organization. It is commonly used in large networks where scalability and manageability are important.
- The primary purpose of tree topology is to create large and complex networks that can span long distances and support multiple devices. It enables better management and control of the network structure

Advantages of Tree Topology

- **Efficient Data Transmission:**
Data traffic can be efficiently managed within specific branches, reducing congestion and improving bandwidth usage especially useful in networks with high data flow.

- **Segmentation:**
It allows logical division of the network into smaller, manageable units based on department, location, or function, enhancing isolation, control, and security.
- **Point-to-Point Connection:**
The root node maintains point-to-point connections with other nodes, offering dedicated bandwidth and minimizing data collisions and [congestion](#).
- **Scalability:**
Tree topology is highly [scalable](#), allowing expansion by adding branches and hierarchical levels to the network.
- **Flexibility:**
It can accommodate diverse sub network types and sizes by using different hubs and cables.
- **Reliability:**
Errors can be easily isolated within the network without affecting the root node, making it more reliable than many other topologies.
- **Security:**
Tree topology enhances both security and privacy by using dedicated channels or links for data transmission, making it suitable for sensitive environments.
- **Centralized Control:**
The structure supports centralized control and management, which simplifies tasks such as monitoring, [troubleshooting](#), and overall security management.

Disadvantages of Tree Topology

- **Single Point of Failure:**
The root node acts as a single point of failure if it crashes or is damaged, it can bring down the entire network.
- **Difficult Maintenance:**
Due to its complex structure, maintenance is not straightforward.
- **Complexity:**
The installation, configuration, and maintenance of this topology can be complex for multi-level networks.
- **Cost:**
The cost associated with setting this topology up is relatively high as we require cables, hubs, and various other network devices for its proper functioning. Hence, the overall cost increases.
- **Dependency:**
In this topology, if the root node and backbone stop working by any chance, then the overall network will be affected by it.
- **Performance degradation:**
Performance degradation, especially in downstream branches, occurs due to heavy network traffic or if the root node becomes overloaded.

Practical Applications of Tree Topology

- **Military Communication Systems:**
Tree topology is utilized in military communication systems to establish secure, structured, and reliable communication links between command centers and various operational units.
- **Aircraft Navigation Systems:**
In aircraft, this topology connects onboard sensors and instruments, allowing seamless data transmission to ground control stations, which supports efficient monitoring and navigation.